

Dimensionality Reduction with Point-distributions Similarity Invariant

Hang Zhang^{1 2} Kai Ming Ting^{1 2}

Abstract

Existing dimensionality reduction methods all perform dimensionality reduction by preserving some invariant in the space before and after dimensionality reduction. This paper proposes a new dimensionality reduction invariant: preserving the invariant of the point-distributions similarity. We also design a linear-time and efficient method to achieve dimensionality reduction while preserving this invariant. We theoretically prove the feasibility of our method for dimensionality reduction. Furthermore, our results on benchmark datasets and single-cell expression data demonstrate the effectiveness and efficiency of the proposed method.

1. Introduction

Dimensionality reduction, as a crucial technique in machine learning, plays a vital role in many fields, serving as a fundamental method for data visualization and preprocessing (Van Der Maaten et al., 2009; Anowar et al., 2021). For example, it is used in atomistic simulations and general analysis of high-dimensional systems in physical chemistry (Rydowski, 2023; Coifman et al., 2008), and as a preprocessing technique in single-cell domains to reduce the dimensionality of high-dimensional gene expression data for subsequent analysis (Shang & Zhou, 2022; Becht et al., 2019; Zhang et al., 2025b).

Existing and widely used dimensionality reduction algorithms, such as Principal Component Analysis (PCA) (Abdi & Williams, 2010), t-Distributed Stochastic Neighbor Embedding (t-SNE) (Maaten & Hinton, 2008), UMAP (McInnes et al., 2018), MultiDimensional Scaling (MDS) (Kruskal & Wish, 1978), and AutoEncoder (AE) (Wang et al., 2016), although very different, share a common char-

acteristic: *they preserve something unchanged (we term this dimensionality reduction invariant) before and after dimensionality reduction*. For example, PCA preserves the dimensions with the maximum variance, t-SNE preserves the probability distribution of point pairs, UMAP preserves the topological structure, and AE preserves the reconstructed variables after dimensionality reduction. Except for PCA, which considers the variables unchanged and is therefore linear, the other algorithms consider the relationships between points unchanged and thus often have a quadratic time complexity (Van Der Maaten et al., 2009).

Existing algorithms often overlook the heterogeneous nature of real-world datasets, implicitly assuming a single generative source. In practice, however, data typically arises from a mixture of k distinct distributions. For example, in classification and clustering datasets, the points in the same class (cluster) can be considered as being generated from the same distribution, k classes (clusters) correspond to k different distributions. Consequently, for downstream tasks such as classification, the preservation of pairwise point-to-point similarities is secondary to maintaining the affinity between a data point and these latent distributions. This observation motivates our central research question: *Is it possible to design a dimensionality reduction technique that explicitly preserves the similarity structure between data points and these distributions?*

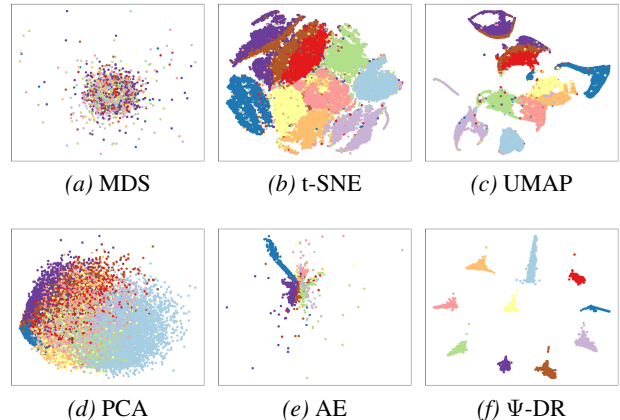


Figure 1. Comparison of dimensionality reduction results on the MNIST dataset. The first row has a complexity of quadratic or higher, while the second row has linear complexity.

¹ State Key Laboratory of Novel Software Technology, Nanjing University, Nanjing 210023, China ²School of Artificial Intelligence, Nanjing University, Nanjing 210023, China. Correspondence to: Kai Ming Ting <tingkm@nju.edu.cn>.

In the absence of label information, we approximate the k underlying distributions by first employing a clustering algorithm to partition the data, followed by Kernel Mean Embedding (Muandet et al., 2017) for distribution estimation. Figure 1 presents a comparative analysis of our proposed dimensionality reduction method, Ψ -DR (or PSIDR), against baseline approaches on the MNIST dataset. By preserving point-distribution similarity invariants, Ψ -DR successfully disentangles the ten distinct digit distributions. Consequently, the resulting embeddings are significantly more conducive to downstream tasks such as classification and clustering.

The main contributions of this paper are summarized as follows:

1. Introducing a novel dimensionality reduction invariant defined by the similarity between points and distributions, rather than traditional point-to-point similarity.
2. Proposing Ψ -DR, an efficient linear-time framework that optimizes dimensionality reduction while maintaining the integrity of this invariant.
3. Validating the effectiveness and efficiency of our method through theoretical analysis and extensive empirical experiments on benchmark datasets and single-cell expression data.

2. Related Work

Dimensionality reduction techniques aim to map high-dimensional data into a lower-dimensional manifold while preserving specific dimensionality reduction invariants. Based on the nature of the preserved invariant, we categorize existing literature into three main streams: variance-based, pairwise-similarity-based, and reconstruction-based invariant methods.

The most fundamental invariant in classical statistical analysis is the global variance, which represents the information content of the data. PCA (Pearson, 1901) is the pioneer of this category, preserving the maximum variance via orthogonal projection. Many other variants of PCA have been proposed to address the interpretation difficulty of PCA’s dense loading vectors (Zou et al., 2006), to overcome the linearity limitation of PCA (Schölkopf et al., 1997), and to be robust to outliers (Candès et al., 2011).

A vast body of research focuses on preserving the geometric relationships between data points. These methods assume that the data lies on a low-dimensional manifold embedded in the high-dimensional space. MDS (Cox & Cox, 2008) preserves global Euclidean distances. Isomap (Tenenbaum et al., 2000) extends MDS by replacing Euclidean distances with geodesic distances to capture the

true manifold geometry. On the other hand, Locally Linear Embedding (LLE) (Roweis & Saul, 2000) and Laplacian Eigenmaps (Belkin & Niyogi, 2003) focus on local invariants. t-SNE (Maaten & Hinton, 2008) minimizes the Kullback-Leibler divergence between joint probabilities in high and low dimensions, excelling at visualization but struggling with global structure. UMAP (McInnes et al., 2018) improves upon this by utilizing Riemannian geometry and fuzzy simplicial sets to better balance local and global structure preservation.

With the rise of deep learning, preserving the reconstruction capability, the ability to recover the original input from the latent space, has become a dominant paradigm. The standard Autoencoder (Hinton & Salakhutdinov, 2006) learns a compressed representation by minimizing the reconstruction error. Many variations were proposed afterwards (Vincent et al., 2008; Kingma & Welling, 2013; Moor et al., 2020).

Despite the diversity of these methods, they predominantly focus on either global statistics, pairwise point-to-point relations ($x_i \leftrightarrow x_j$), or self-reconstruction ($x_i \leftrightarrow \hat{x}_i$). This paper proposes a new invariant based on the similarity between points and k distributions ($x_i \leftrightarrow \mathcal{P}_k$) in the data.

3. Problem Formulation

Let $\mathcal{X} \subset \mathbb{R}^d$ be a compact input space, and let $\mathcal{P}(\mathcal{X})$ denote the set of Borel probability measures on $\mathcal{X}$. Suppose the observed data $\{x_i\}_{i=1}^n \subset \mathcal{X}$ are generated from a mixture of k underlying probability distributions $\{\nu_c\}_{c=1}^k \subset \mathcal{P}(\mathcal{X})$, i.e.,

$$x_i \sim \nu_{c(i)}, \quad c(i) \in \{1, \dots, k\}.$$

Our goal is to learn a low-dimensional embedding $z_i \in \mathbb{R}^p$ ($p \ll d$) for each x_i such that the **similarity structure among the underlying distributions $\{\nu_c\}$ and between points and distributions** is faithfully preserved.

Formally, let $\rho : \mathcal{P}(\mathcal{X}) \times \mathcal{P}(\mathcal{X}) \rightarrow [0, 1]$ be a similarity function satisfying $\rho(\nu, \nu') = \langle \nu, \nu' \rangle_{\mathcal{H}}$ for the Hilbert space $\mathcal{H}$, and let $\sigma : \mathcal{X} \times \mathcal{P}(\mathcal{X}) \rightarrow [0, 1]$ denote the point-distribution similarity, defined as $\sigma(x, \nu) = \mathbb{E}_{x' \sim \nu}[\kappa(x, x')]$ for some symmetric positive definite kernel κ . We seek an embedding map $f : \mathcal{X} \rightarrow \mathbb{R}^p$ and a set of centers $\{\bar{z}_c\}_{c=1}^k \subset \mathbb{R}^p$ such that for all i and c ,

$$\begin{aligned} \langle f(x_i), \bar{z}_c \rangle_{\mathbb{R}^p} &\approx \sigma(x_i, \nu_c), \\ \text{and } \|\bar{z}_c - \bar{z}_{c'}\|_2^2 &\approx \|\nu_c - \nu_{c'}\|_{\mathcal{H}}^2. \end{aligned}$$

This leads to the following **point-distributions similarity invariant-preserving dimensionality reduction** problem:

Problem 3.1. Given the dataset $\{x_i\}_{i=1}^n$ and the set of underlying distributions $\{\nu_c\}_{c=1}^k$, our objective is to find

an embedding map $f : \mathcal{X} \rightarrow \mathbb{R}^p$ and a set of centers $\{\bar{z}_c\}_{c=1}^k \subset \mathbb{R}^p$ (where $p \ll d$) that minimize the following objective function:

$$\min_{f, \{\bar{z}_c\}} \mathcal{L}(f, \{\bar{z}_c\}) = \sum_{i=1}^n \sum_{c=1}^k (\langle f(x_i), \bar{z}_c \rangle - \sigma(x_i, \nu_c))^2 + \lambda \sum_{c, c'=1}^k (\|\bar{z}_c - \bar{z}_{c'}\|_2^2 - \|\nu_c - \nu_{c'}\|_{\mathcal{H}}^2)^2,$$

subject to $p \ll d$, where $\lambda > 0$ balances point-distribution and inter-distribution similarity preservation.

However, Problem 3.1 is intractable in practice: the true distributions $\{\nu_c\}$ are unknown, and the similarity functionals σ and $\|\cdot\|_{\mathcal{H}}$ cannot be evaluated without access to the full distributions.

To render the problem computable, we represent each distribution ν_c via its **kernel mean embedding** (KME) in a reproducing kernel Hilbert space (RKHS) (Berlinet & Thomas-Agnan, 2011).

Let $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be a continuous, bounded, positive definite kernel with RKHS $\mathcal{H}_\kappa$ and feature map $\phi : \mathcal{X} \rightarrow \mathcal{H}_\kappa$, such that $\kappa(x, x') = \langle \phi(x), \phi(x') \rangle_{\mathcal{H}_\kappa}$. The kernel mean embedding of ν_c is defined as: $\mu_c := \mathbb{E}_{x \sim \nu_c}[\phi(x)] \in \mathcal{H}_\kappa$.

Under mild conditions (e.g., κ being characteristic), the map $\nu \mapsto \mu_\nu$ is injective, ensuring that distributional information is preserved in $\mathcal{H}_\kappa$.

Given empirical samples $\{x_i\}_{i \in C_c}$ from subset C_c that drawn from ν_c , we approximate μ_c by the empirical mean embedding: $\hat{\mu}_c := \frac{1}{|C_c|} \sum_{i \in C_c} \phi(x_i)$.

Crucially, the point-distribution similarity becomes

$$\sigma(x_i, \nu_c) \approx \langle \phi(x_i), \hat{\mu}_c \rangle_{\mathcal{H}_\kappa} = \frac{1}{|C_c|} \sum_{j \in C_c} k(x_i, x_j),$$

and the squared distance between distributions is

$$\begin{aligned} \|\nu_c - \nu_{c'}\|_{\mathcal{H}_\kappa}^2 &\approx \|\hat{\mu}_c - \hat{\mu}_{c'}\|_{\mathcal{H}_\kappa}^2 \\ &= \langle \hat{\mu}_c, \hat{\mu}_c \rangle + \langle \hat{\mu}_{c'}, \hat{\mu}_{c'} \rangle - 2\langle \hat{\mu}_c, \hat{\mu}_{c'} \rangle. \end{aligned}$$

Substituting these empirical estimates into Problem 3.1, we obtain the following **kernel mean embedding-based similarity-preserving DR** problem:

Problem 3.2. Let $\{C_c\}_{c=1}^k$ be a subset of $\{1, \dots, n\}$. Find $Z = [z_1, \dots, z_n]^\top \in \mathbb{R}^{n \times p}$ and $\{\bar{z}_c\}_{c=1}^k \subset \mathbb{R}^p$ that minimize

$$\begin{aligned} \mathcal{L}_{\text{KME}}(Z, \{\bar{z}_c\}) &= \sum_{i=1}^n \sum_{c=1}^k (\langle z_i, \bar{z}_c \rangle - \langle \phi(x_i), \hat{\mu}_c \rangle_{\mathcal{H}_\kappa})^2 \\ &\quad + \lambda \sum_{c, c'=1}^k (\|\bar{z}_c - \bar{z}_{c'}\|_2^2 - \|\hat{\mu}_c - \hat{\mu}_{c'}\|_{\mathcal{H}_\kappa}^2)^2. \end{aligned}$$

Problem 3.2 is now fully data-driven and computable via kernel function. The proposed algorithm solves Problem 3.2 through a decoupled two-stage procedure:

- (i) embedding distribution via MDS on $\|\hat{\mu}_c - \hat{\mu}_{c'}\|_{\mathcal{H}_\kappa}$.
- (ii) localizing points via multilateration using distances derived from $\langle \phi(x_i), \hat{\mu}_c \rangle_{\mathcal{H}_\kappa}$.

4. Optimization Problem Decomposition and Solution Strategy

The joint optimization Problem 3.2 is non-convex and high-dimensional. We adopt a **decoupled two-stage strategy** that first embeds the distributional prototypes and then localizes the data points with respect to these fixed prototypes. This approach is not only computationally efficient but also theoretically well-motivated, as it directly controls the primary objective of inter-distribution similarity preservation before addressing point-level fidelity.

Since Problem 3.2 requires dividing the data into k subsets, to ensure the effectiveness and efficiency of the division, we use the recently proposed linear clustering method KBC (Zhang et al., 2025a) to obtain k subsets. Then, our method decomposes Problem 3.2 into the following sequential sub-problems:

Embedding Distributional Prototypes

We first isolate the inter-distribution term in $\mathcal{L}_{\text{KME}}$. Since the empirical kernel mean embeddings $\{\hat{\mu}_c\}_{c=1}^k$ are now fixed, the problem of finding their low-dimensional counterparts $\{\bar{z}_c\}_{c=1}^k$ reduces to a **classical multidimensional scaling (MDS)** problem.

From the kernel mean embeddings, we compute the empirical squared MMD (Maximum Mean Discrepancy) matrix $\mathbf{D} \in \mathbb{R}^{k \times k}$, where $D_{cc'} = \|\hat{\mu}_c - \hat{\mu}_{c'}\|_{\mathcal{H}_\kappa}^2$. Classical MDS seeks a configuration of points $\{\bar{z}_c\}_{c=1}^k \subset \mathbb{R}^p$ whose Euclidean distances best match $\mathbf{D}$. This is achieved by minimizing the strain:

$$\min_{\{\bar{z}_c\} \subset \mathbb{R}^p} \sum_{c, c'=1}^k (\|\bar{z}_c - \bar{z}_{c'}\|_2^2 - D_{cc'})^2. \quad (1)$$

This problem admits a closed-form solution via an eigen-decomposition of the double-centered distance matrix, ensuring that the global structure of the distributional space is preserved in $\mathbb{R}^p$.

Localizing Data Points via Multilateration

With the low-dimensional centers $\{\bar{z}_c\}_{c=1}^k$ fixed from Stage 1, we now solve for the individual point embeddings $\{z_i\}_{i=1}^n$. We first convert the point-distribution inner products from Problem 3.2 into distances, which is more amenable to localization.

The point-distribution similarity $\langle \phi(x_i), \hat{\mu}_c \rangle_{\mathcal{H}_\kappa}$ can be converted to an approximate squared distance in the embedding space using: $d_{ic}^2 \approx 2(1 - \langle \phi(x_i), \hat{\mu}_c \rangle_{\mathcal{H}_\kappa})$, assuming that the embeddings are normalized¹. The problem for each point z_i becomes to find its location such that its distances to the fixed anchors $\{\bar{z}_c\}$ match the target distances $\{d_{ic}\}$.

This is the well-known **multilateration problem** (Widdison & Long, 2024; Norrdine, 2012):

$$z_i = \arg \min_{z \in \mathbb{R}^p} \sum_{c=1}^k (\|z - \bar{z}_c\|_2 - d_{ic})^2. \quad (2)$$

For $k > p + 1$, this is an over-determined system. We use the closed-form least-squares solution derived from squaring the distance equations and canceling quadratic terms. This yields a linear system $Az = b_i$, which can be solved efficiently via the normal equations. In degenerate cases (e.g., anchors are co-planar in a way that makes A rank-deficient), we fall back to a robust numerical optimizer².

Post-processing: Subset Center Alignment

The two-stage procedure does not guarantee that the empirical mean of the embedded points in subset c will coincide with its target center $\bar{z}_c$. To enforce this consistency post-hoc, we perform a final rigid translation for each subset:

$$z_i \leftarrow z_i + (\bar{z}_c - \frac{1}{|C_c|} \sum_{j \in C_c} z_j), \quad \forall i \in C_c.$$

This step is optional in practice, as it is not a crucial part of maintaining dimensionality reduction invariant (point-distribution similarity).

This pipeline provides a highly efficient and effective approximate solution to the intractable joint Problem 3.2, with each stage being either closed-form or involving a simple, well-conditioned optimization.

The proposed Ψ -DR³ is shown in Algorithm 1.

Time Complexity Analysis: Let n be the number of data points, k be the number of distributions, and p be the target dimension ($p \ll d$). Denote $\bar{c} = \max_c |C_c|$ as the size of the largest subset.

We analyze the time complexity of the proposed Ψ -DR algorithm across its four stages. The preprocessing phase, comprising KBC clustering and the computation of k kernel mean embeddings, incurs a cost of $O(nkd) + O(\bar{c})$.

¹Throughout this paper, d_{ic} denotes the scalar target distance between the i -th data point and the c -th distribution center. We define $\mathbf{d}_i = [d_{i1}, \dots, d_{ik}]^\top \in \mathbb{R}^k$ as the vector collecting these distances for point i . In the theoretical analysis (e.g., Lemma 5.6), we drop the index i and use $\mathbf{d}$ to denote a generic distance vector.

²See details in Appendix.

³The source code is available in <https://github.com/IsolationKernel/PSIDR>.

Algorithm 1 Ψ -DR (or PSIDR): Point-distributions Similarity Invariant Dimensionality Reduction

Require: Data $\mathcal{X} \in \mathbb{R}^{n \times d}$, number of distributions k , target dimension p , kernel $k(\cdot, \cdot)$.

Ensure: Embedding $\mathbf{Z} \in \mathbb{R}^{n \times p}$

- 1: **Obtain subset and kernel mean embedding**
 - 2: $\mathbf{c} \leftarrow \text{KBC}(\mathcal{X}; k)$ {Obtain subset $\{C_c\}$ }
 - 3: $\hat{\mu}_c = \frac{1}{|C_c|} \sum_{i \in C_c} \phi(x_i), \quad \forall c = 1, \dots, k$
 - 4: **Embedding distributional prototypes**
 - 5: $\mathbf{D} \leftarrow D_{cc'} = \|\hat{\mu}_c - \hat{\mu}_{c'}\|_{\mathcal{H}_\kappa}^2$
 - 6: $\bar{\mathbf{Z}} \leftarrow \text{MDS}(\mathbf{D}, p)$
 - 7: **Localizing data points (Multilateration)**
 - 8: **for** $i = 1$ **to** n **do**
 - 9: $d_{ic} = \sqrt{2(1 - \langle \phi(x_i), \hat{\mu}_c \rangle)}$
 - 10: $z_i \leftarrow \arg \min_{z \in \mathbb{R}^p} \sum_{c=1}^k (\|z - \bar{z}_c\|_2 - d_{ic})^2$
 - 11: **end for**
 - 12: **Post-processing (optional)**
 - 13: **for** $c = 1$ **to** k **do**
 - 14: $\hat{\mu}_c \leftarrow \frac{1}{|C_c|} \sum_{j \in C_c} z_j$
 - 15: $z_i \leftarrow z_i + (\bar{z}_c - \hat{\mu}_c), \quad \forall i \in C_c$
 - 16: **end for**
 - 17: **Return** $\mathbf{Z}$
-

Subsequently, constructing the pairwise MMD matrix $\mathbf{D}$ requires $O(k^2)$ operations. The embedding of distributional prototypes relies on classical MDS, which involves an eigendecomposition with a complexity of $O(k^3)$. In the localization phase, computing target distances takes $O(nk)$, while solving the closed-form multilateration problem for n points scales as $O(np^2k)$. Finally, the post-processing adds a negligible cost of $O(np)$. Consequently, the overall time complexity is $O(k^3 + np^2k + nkd)$. Since the number of distributions k and the target dimension p are typically much smaller than the sample size n ($k, p \ll n$), the algorithm achieves linear scalability with respect to n and d .

5. Theoretical Analysis

We now provide a statistical analysis of Algorithm 1. Our main result (Theorem 5.7) establishes that the learned embedding is *consistent* in the sense that it converges in mean squared error to an ideal similarity-preserving configuration as the sample size grows. The proof proceeds through three key lemmas that carefully quantify estimation error, metric distortion, and localization sensitivity. The lemmas and theorems are summarized in Table 1.

We begin by stating and motivating our assumptions.

Assumption 5.1 (Bounded and Characteristic Kernel). The kernel $\kappa : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is continuous, positive definite, and bounded: there exists $\varepsilon > 0$ such that $\sup_{x \in \mathcal{X}} \kappa(x, x) \leq \varepsilon$. Moreover, κ is characteristic, so the mean embedding map $\nu \mapsto \mu_\nu = \mathbb{E}_{x \sim \nu}[\phi(x)]$ is injective.

Table 1. Summary of theoretical results for Distributional similarity preserved dimensionality reduction.

Result	Description	Reference
KME Concentration	Empirical KME converge uniformly to true means at rate $O(1/\sqrt{n_{\min}})$.	Lem. 5.4
MDS Stability	Center embedding is stable under distance matrix perturbations.	Lem. 5.5
Multiliteration Sensitivity	Point localization is Lipschitz continuous in distances and anchors.	Lem. 5.6
Embedding Consistency	Learned embedding converges to ideal configuration as $n \rightarrow \infty$.	Thm. 5.7
Finite-Sample Error Bound	Non-asymptotic bound on average embedding error.	Thm. 5.8
Distance Preservation	Point-to-distribution distances in $\mathbb{R}^p$ consistently estimate RKHS distances.	Thm. 5.9

This assumption is standard in kernel methods and ensures that the feature map $\phi(x)$ has uniformly bounded norm, which is crucial for concentration inequalities. The characteristic property guarantees that distinct distributions have distinct embeddings, which is fundamental to preserving distributional information in the RKHS. Without this, different distributions could be mapped to the same point in $\mathcal{H}_\kappa$, making similarity preservation impossible.

Assumption 5.2 (Asymptotically Balanced subset). There exists a partition $\{C_c\}_{c=1}^k$ of $\{1, \dots, n\}$ such that $n_{\min} := \min_{c \in [k]} |C_c| \rightarrow \infty$ as $n \rightarrow \infty$.

This assumption ensures that every subset receives sufficient data to accurately estimate its kernel mean embedding. It rules out pathological cases where a subset contains only a constant number of points, which would lead to inconsistent KME estimates. In practice, clustering algorithms like KBC (used in our method) tend to produce reasonably balanced subsets under mild data regularity conditions, making this assumption realistic for large datasets.

Assumption 5.3 (Non-degenerate Limit Geometry). Let $\mathbf{D}^* \in \mathbb{R}^{k \times k}$ with $D_{cc'}^* = \|\mu_c - \mu_{c'}\|_{\mathcal{H}_\kappa}^2$, and let $\mathbf{B}^* = -\frac{1}{2}\mathbf{H}\mathbf{D}^*\mathbf{H}$, where $\mathbf{H} = \mathbf{I}_k - \frac{1}{k}\mathbf{1}_k\mathbf{1}_k^\top$. Denote the eigenvalues of $\mathbf{B}^*$ by $\lambda_1^* \geq \dots \geq \lambda_k^* \geq 0$. Assume that for the target dimension $p < k$, the eigengap $\Delta_p := \lambda_p^* - \lambda_{p+1}^*$ satisfies $\Delta_p > 0$. Furthermore, assume the matrix of true MDS centers $\bar{\mathbf{Z}}^* \in \mathbb{R}^{k \times p}$ has full column rank.

This geometric assumption ensures that the true low-dimensional configuration of centers is identifiable and stable. A positive eigengap Δ_p guarantees that the top- p eigenspace of the Gram matrix is well-separated from the rest, which is a standard requirement for the consistency of spectral methods like MDS (see, e.g., Von Luxburg et al. (2008)). The full-rank condition on $\bar{\mathbf{Z}}^*$ prevents degenerate configurations (e.g., all centers lying on a lower-dimensional subspace), which would make the multiliteration problem ill-posed.

With these assumptions in place, we now establish the key technical lemmas.

Lemma 5.4 (Uniform Concentration of Kernel Mean Embeddings). Suppose Assumption 5.1 holds. For any $\delta \in$

$(0, 1)$, with probability at least $1 - \delta$,

$$\max_{c \in [k]} \|\hat{\mu}_c - \mu_c\|_{\mathcal{H}_\kappa} \leq \sqrt{\frac{\varepsilon}{n_{\min}}} (1 + \sqrt{6 \log \frac{2k}{\delta}}).$$

Lemma 5.4 quantifies the uniform estimation error of the empirical kernel mean embeddings across all k distributions. The $O(1/\sqrt{n_{\min}})$ rate is minimax optimal and matches the parametric rate for mean estimation in Hilbert spaces. The logarithmic dependence on k arises from the union bound and is unavoidable when controlling the error uniformly over multiple distributions.

Lemma 5.5 (Perturbation Bound for MDS Embeddings). Let $\mathbf{D}, \mathbf{D}^* \in \mathbb{R}^{k \times k}$ be Euclidean distance matrices, and let $\bar{\mathbf{Z}}, \bar{\mathbf{Z}}^* \in \mathbb{R}^{k \times p}$ be their classical MDS embeddings (i.e., $\bar{\mathbf{Z}}\bar{\mathbf{Z}}^\top = \mathbf{B}$, where $\mathbf{B} = -\frac{1}{2}\mathbf{H}\mathbf{D}\mathbf{H}$, and similarly for $\bar{\mathbf{Z}}^*$). Let $\mathcal{O}(p)$ denote the set of $p \times p$ orthogonal matrices. Under Assumption 5.3,

$$\min_{\mathbf{Q} \in \mathcal{O}(p)} \|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}^*\mathbf{Q}\|_F \leq \frac{\sqrt{2p}}{2\Delta_p} \|\mathbf{D} - \mathbf{D}^*\|_F.$$

Lemma 5.5 formalizes the stability of the MDS step: small perturbations in the input distance matrix lead to small perturbations in the output embedding, up to an orthogonal transformation. The bound depends inversely on the eigengap Δ_p , highlighting that well-separated eigenvalues (a non-degenerate spectrum) are crucial for stability. This result bridges the statistical error from Lemma 5.4 (which affects $\mathbf{D}$) to geometric error in the centers $\bar{\mathbf{Z}}$.

Lemma 5.6 (Multiliteration Sensitivity). Let $\bar{\mathbf{Z}} = [\bar{z}_1, \dots, \bar{z}_k]^\top \in \mathbb{R}^{k \times p}$ have full column rank, and define $\mathbf{A} = -2(\bar{\mathbf{Z}}_{2:k} - \mathbf{1}_{k-1}\bar{z}_1^\top) \in \mathbb{R}^{(k-1) \times p}$. Assume $\sigma_{\min}(\mathbf{A}) \geq \sigma_0 > 0$. Let $\mathbf{d} \in \mathbb{R}^k$ be distances, and define $\mathbf{b} \in \mathbb{R}^{k-1}$ by $b_c = d_{c+1}^2 - d_1^2 - (\|\bar{z}_{c+1}\|_2^2 - \|\bar{z}_1\|_2^2)$. Let $z^\dagger = \mathbf{A}^\dagger \mathbf{b}$ ⁴. Then for any two inputs $(\mathbf{d}, \bar{\mathbf{Z}})$ and $(\mathbf{d}', \bar{\mathbf{Z}}')$,

$$\|z^\dagger - z'^\dagger\|_2 \leq L \left(\|\mathbf{d}^2 - \mathbf{d}'^2\|_2 + \|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}'\|_F \right),$$

where the Lipschitz constant $L = O\left(\frac{1}{\sigma_0^2} \left(1 + \frac{\|\mathbf{b}\|_2}{\sigma_0}\right)\right)$.

⁴ denotes the Moore-Penrose pseudoinverse.

This lemma captures the robustness of the closed-form multilateration procedure. The constant L depends on the smallest singular value σ_0 of the anchor matrix $\mathbf{A}$, which measures how well-conditioned the localization problem is. When anchors are in general position (as ensured by Assumption 5.3), σ_0 is bounded away from zero, making the localization stable. This result ensures that errors in the target distances (from KME estimation) and anchor positions propagate linearly to the final point embeddings.

Theorem 5.7 (Consistency of the Embedding). *Let $z_i^* \in \mathbb{R}^p$ satisfy $\|z_i^* - \bar{z}_c^*\|_2^2 = 2(1 - \langle \phi(x_i), \mu_c \rangle_{\mathcal{H}_\kappa})$ for all c . Let z_i be the embedding. Then, as $n \rightarrow \infty$,*

$$\frac{1}{n} \sum_{i=1}^n \|z_i - z_i^*\|_2^2 \xrightarrow{\mathbb{P}} 0.$$

Theorem 5.7 states that the embedding produced by our algorithm converges to an ideal configuration that perfectly preserves the distributional similarities defined by the kernel mean embeddings. The proof combines the three lemmas in a clean error propagation argument: KME estimation error (Lemma 5.4) induces error in the MDS step (Lemma 5.5), which in turn affects point localization (Lemma 5.6). The rate of convergence is $O(k^2/n_{\min})$, which is favorable when the number of distributions k is fixed and subsets are balanced ($n_{\min} = \Theta(n/k)$).

Theorem 5.8 (Finite-Sample Embedding Error Bound). *Under Assumptions 5.1, 5.2, and 5.3, there exists a constant $C > 0$, such that for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\frac{1}{n} \sum_{i=1}^n \|z_i - z_i^*\|_2^2 \leq C \cdot \frac{k^2}{n_{\min}} \left(1 + \log \frac{2k}{\delta}\right).$$

While Theorem 5.7 establishes asymptotic consistency, Theorem 5.8 provides a more practical guide by characterizing the convergence rate. Two key insights emerge from this bound: (i) The error bound scales inversely with $n_{\min}$, not the total sample size n . This highlights that the performance of Ψ -DR is determined by the least represented distribution. To achieve a low embedding error, the dataset should ideally be balanced; for highly imbalanced data, $n_{\min}$ becomes the bottleneck. (ii) The error scales quadratically with the number of distributions k . This reflects the cumulative uncertainty: as k increases, the MDS stage (embedding k centers) accumulates more estimation errors from the $k \times k$ MMD matrix, which subsequently propagate to the multilateration stage.

Theorem 5.9 (Point-to-Distribution Distance Preservation). *Assume the kernel is normalized ($k(x, x) = 1$ for all x). Let $d_{ic}^{\mathcal{H}} = \|\phi(x_i) - \mu_c\|_{\mathcal{H}_\kappa}$ and $d_{ic}^{\mathbb{R}^p} = \|z_i - \bar{z}_c\|_2$. Under Assumptions 5.1, 5.2, and 5.3, for any $\delta \in (0, 1)$,*

exist $C' > 0$, with probability at least $1 - \delta$,

$$\max_{i \in [n], c \in [k]} |d_{ic}^{\mathbb{R}^p} - d_{ic}^{\mathcal{H}}| \leq C' \cdot \frac{k}{\sqrt{n_{\min}}} (1 + \sqrt{\log \frac{2k}{\delta}}).$$

This result is of central importance as it directly validates the core motivation of our method. Recall that our objective (Problem 3.2) is to preserve the similarity between points and distributions. Theorem 5.9 guarantees that, with high probability, the Euclidean distances $\|z_i - \bar{z}_c\|_2$ in the low-dimensional space $\mathbb{R}^p$ are consistent estimates of the abstract RKHS distances $\|\phi(x_i) - \mu_c\|_{\mathcal{H}_\kappa}$. This metric preservation property implies that downstream tasks, such as k -means, perform comparably in the linear embedding space as they would using the computationally expensive kernel methods in the original high-dimensional space. The convergence rate of $O(1/\sqrt{n_{\min}})$ is consistent with standard results for kernel mean estimation, indicating that our two-stage reduction procedure does not degrade the statistical efficiency of the underlying estimator.

6. Experiments

To demonstrate the effectiveness of our proposed Ψ -DR algorithm, we compared it with existing algorithms, PCA, t-SNE, UMAP, AE, and MDS on benchmark datasets and single-cell gene expression datasets. These datasets range in size from 1372 to 804414 samples and have dimensions from 3 to 47236. We set the target dimension to 2. Detailed information about the datasets is provided in the appendix.

To evaluate intrinsic clustering structure and compactness without reliance on downstream tasks, we use the Davies–Bouldin Index (DB) (Davies & Bouldin, 2009) and the Calinski–Harabasz Index (CH) (Caliński & Harabasz, 1974). Lower DB scores and higher CH scores indicate better separation and cohesion of clusters, reflecting more effective dimensionality reduction. Since reduced representations are often used in downstream applications, we further assess their utility through supervised and unsupervised tasks. We perform k -means clustering on the embeddings and report Normalized Mutual Information (NMI) (Vinh et al., 2010) between predicted clusters and ground-truth labels. Additionally, we evaluate classification performance by training a random forest classifier (Breiman, 2001) on an 80% training split of the reduced data and reporting accuracy on the remaining 20% test split.

6.1. Comparison between different invariants

The result of the dimensionality reduction comparison is shown in Table 2. Ψ -DR outperforms other methods in terms of CH, DB, and NMI. t-SNE achieved the best results in terms of ACC.

We summarize the number of winners and runner-ups for

Table 2. Dimensionality Reduction Performance Comparison. The winner is shown in bold, the runner-up in underlined.

Dataset	Calinski-Harabasz ($\uparrow$)						Davies-Bouldin ($\downarrow$)					
	PCA	t-SNE	UMAP	AE	MDS	Ψ -DR	PCA	t-SNE	UMAP	AE	MDS	Ψ -DR
COIL20	298	1284	4058	87	90	<u>3985</u>	6.84	<u>3.82</u>	4.01	4.99	8.54	0.72
Banknote	298	792	<u>1369</u>	562	342	37146	1.92	1.15	<u>0.84</u>	1.26	1.76	0.05
Landsat	1158	1244	<u>1609</u>	1118	826	1990	1.27	<u>1.08</u>	1.18	1.19	1.28	1.06
Pendigits	2270	4078	<u>6528</u>	1306	2092	7470	5.74	1.97	<u>1.53</u>	4.78	4.15	1.27
USPS	1063	5286	9756	726	652	<u>8252</u>	5.62	1.77	<u>1.42</u>	18.93	8.75	1.05
Letters	339	660	193	388	344	<u>418</u>	20.34	<u>13.73</u>	17.66	19.31	27.34	11.92
MNIST	4986	30448	<u>46927</u>	3246	2	52306	8.47	1.48	<u>1.46</u>	5.51	191.87	0.96
Skin	33589	<u>59854</u>	39187	2632	NA	1279417	1.65	<u>1.33</u>	1.60	2.03	NA	0.13
Gisette	689	4911	<u>7346</u>	865	1	21224	2.70	1.07	<u>0.89</u>	1.83	102.46	0.34
Rcv1	<u>7199</u>	5281	NA	NA	NA	15722	73.54	<u>27.69</u>	NA	NA	NA	15.73

Dataset	Accuracy ($\uparrow$)						Normalized Mutual Information ($\uparrow$)					
	PCA	t-SNE	UMAP	AE	MDS	Ψ -DR	PCA	t-SNE	UMAP	AE	MDS	Ψ -DR
COIL20	0.729	<u>0.941</u>	0.865	0.896	0.795	0.997	0.626	<u>0.864</u>	0.839	0.622	0.626	0.960
Banknote	0.876	1.000	<u>0.996</u>	0.989	0.935	<u>0.996</u>	0.011	0.349	<u>0.793</u>	0.646	0.011	0.936
Landsat	0.835	0.900	0.865	<u>0.868</u>	0.843	0.855	0.537	<u>0.634</u>	0.618	0.619	0.542	0.681
Pendigits	0.662	0.996	<u>0.992</u>	0.968	0.806	0.906	0.545	<u>0.811</u>	0.795	0.600	0.581	0.819
USPS	0.379	0.952	<u>0.910</u>	0.873	0.372	0.825	0.309	0.673	<u>0.741</u>	0.414	0.256	0.799
Letters	0.242	0.944	<u>0.887</u>	0.654	0.258	0.607	0.186	<u>0.488</u>	0.437	0.259	0.201	0.513
MNIST	0.321	<u>0.938</u>	0.911	0.743	0.198	0.970	0.209	0.671	<u>0.717</u>	0.304	0.075	0.814
Skin	0.995	0.999	0.999	0.997	NA	0.996	0.014	0.006	<u>0.202</u>	0.005	NA	0.828
Gisette	0.644	0.961	<u>0.954</u>	0.937	0.525	0.931	0.070	0.630	0.702	0.149	0.001	<u>0.655</u>
Rcv1	0.384	0.624	NA	NA	NA	<u>0.547</u>	<u>0.148</u>	0.128	NA	NA	NA	0.343

Table 3. Summary of the Comparison.

Method	PCA	t-SNE	UMAP	AE	MDS	Ψ -DR
winner	0	9	4	0	0	28
runner-up	1	12	19	1	0	6

each algorithm in Table 3, with Ψ -DR achieving the most winner counts. This demonstrates the effectiveness of our proposed method of using point and distribution similarity as a dimensionality reduction invariant, and the effectiveness of the Ψ -DR algorithm. This effectiveness is reflected not only in the dimensionality reduction performance using CH and DB as evaluation metrics, but also in the effectiveness of downstream tasks such as unsupervised k -means clustering using NMI as an evaluation metric.

Table 4 shows the running time of different algorithms. PCA, as a well-known linear-time dimensionality reduction method, is very efficient in dimensionality reduction. Our proposed Ψ -DR, as a linear-time dimensionality reduction algorithm, is also very efficient, second only to PCA. It is worth noting that on the Rcv1 dataset, which has a very large number of points (804,414) and a very high original dimension (47,236), Ψ -DR is more efficient because it is

Table 4. Computation Time (seconds). For times exceeding 100 seconds, we report integer values.

Dataset	PCA	t-SNE	UMAP	AE	MDS	Ψ -DR
COIL20	0.06	6.52	15.65	4.62	40.06	<u>0.66</u>
Banknote	0.01	6.53	3.01	3.90	14.12	<u>0.15</u>
Landsat	0.01	10.82	5.66	5.30	72.47	<u>0.16</u>
Pendigits	0.01	84.54	26.93	28.71	3437	<u>2.19</u>
USPS	0.03	81.90	15.13	28.24	3418	<u>6.32</u>
Letters	0.01	180	15.51	52.11	12110	<u>10.47</u>
MNIST	1.05	1030	117	184	19472	<u>28.55</u>
Skin	0.06	2207	1518	612	NA	<u>22.71</u>
Gisette	0.47	10.36	61.29	48.42	324	<u>2.92</u>
Rcv1	<u>549</u>	35638	NA	NA	NA	382

linear with respect to the number of points and the original dimension, while PCA has the least quadratic complexity with respect to the original dimension.

6.2. Comparison on single-cell expression data

Dimensionality reduction algorithms are widely used in single-cell expression data, which often have very high dimensionality (Tsuyuzaki et al., 2020; Song et al., 2022; Wu

Table 5. Clustering Performance Comparison on single-cell expression data. The winner is shown in bold, the runner-up in underlined.

Dataset	Calinski-Harabasz ($\uparrow$)						Davies-Bouldin ($\downarrow$)					
	PCA	t-SNE	UMAP	AE	MDS	Ψ -DR	PCA	t-SNE	UMAP	AE	MDS	Ψ -DR
airway	9410	607	2564	1502	5	<u>3727</u>	<u>2.62</u>	5.92	3.05	1.29	25.43	2.13
crohn	13862	726	515	1196	2	<u>8407</u>	<u>6.52</u>	15.15	20.61	5.58	180.94	6.50
tonsil	1895	75	170	<u>757</u>	2	719	<u>7.15</u>	19.97	16.20	3.35	108.75	15.62
jurkat	<u>1661</u>	72	16	405	9	4184	<u>0.13</u>	0.42	0.95	0.44	4.38	0.04

Dataset	Accuracy ($\uparrow$)						Normalized Mutual Information ($\uparrow$)					
	PCA	t-SNE	UMAP	AE	MDS	Ψ -DR	PCA	t-SNE	UMAP	AE	MDS	Ψ -DR
airway	0.890	0.762	0.773	0.951	0.727	<u>0.907</u>	<u>0.604</u>	0.224	0.281	0.379	0.130	0.660
crohn	0.583	0.557	0.393	0.761	0.227	<u>0.607</u>	0.466	0.272	0.220	0.511	0.068	<u>0.490</u>
tonsil	0.559	0.351	0.330	0.840	0.360	<u>0.577</u>	0.455	0.107	0.154	0.547	0.158	<u>0.424</u>
jurkat	1.000	1.000	0.994	0.997	0.987	1.000	0.001	0.025	0.016	<u>0.035</u>	0.001	0.198

Table 6. Computation Time on single-cell expression data.

Dataset	PCA	t-SNE	UMAP	AE	MDS	Ψ -DR
airway	0.17	13.34	49.92	21.63	224.77	<u>1.66</u>
crohn	0.60	109.57	108.11	106.07	6540.80	<u>2.35</u>
tonsil	0.16	9.30	44.84	17.41	130.01	<u>0.74</u>
jurkat	0.14	2.87	16.95	8.65	11.47	<u>0.16</u>

& Zhang, 2020; Rebuffet et al., 2024). We compare the algorithms on 4 single-cell expression datasets. The results are shown in Table 5. and the runtime (seconds) is shown in Table 6. Ψ -DR and PCA achieved the best results on the dimensionality reduction metrics CH and DB, while Ψ -DR and AE performed better on downstream tasks. Similar to the benchmark data, PCA and Ψ -DR were the most efficient algorithms for dimensionality reduction.

The dimensionality reduction results vary with the number of distributions k are shown in Figure 2. We report the results for different values of k on COIL and MNIST datasets, where COIL contains 20 classes, and MNIST includes 10 classes. In terms of all four metrics, both COIL and MNIST achieved optimal results near the number of distributions corresponding to the true number of classes.

6.3. Comparison with the latest dimensionality reduction algorithms

In this subsection, we compare our method with state-of-the-art dimensionality reduction algorithms that utilize invariants similar to those used in t-SNE and UMAP. Table 7, 8 and 9 show the results comparing our method with Neg-t-SNE (Damrich et al., 2023), InfoNC-t-SNE (Damrich et al., 2023), PaCMAP (Wang et al., 2021), and LocalMAP (Wang et al., 2025). Experimental results demonstrate that the proposed method achieves superior perfor-

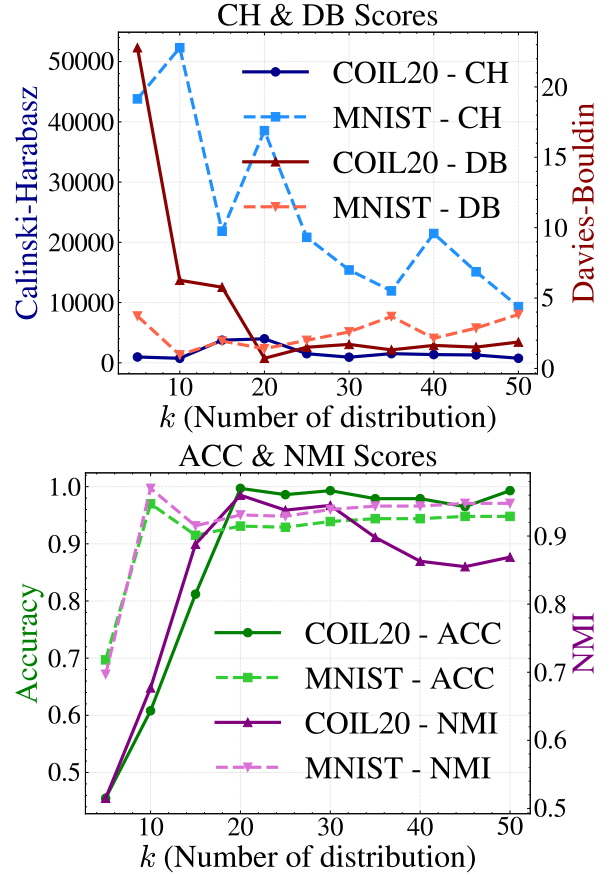


Figure 2. Results of dimensionality reduction vary with the number of distributions.

mance compared to state-of-the-art approaches in terms of both dimensionality reduction quality and efficiency.

We compared the running times on the rcv1 dataset across various random sampling points and dimensions, as shown

Table 7. Dimensionality Reduction Performance Comparison (NtS: Neg-t-SNE, ItS: InfoNC-t-SNE).

Dataset	Calinski-Harabasz ($\uparrow$)					Davies-Bouldin ($\downarrow$)				
	Ψ -DR	NtS	ItS	PaCMAP	LocalMAP	Ψ -DR	NtS	ItS	PaCMAP	LocalMAP
COIL20	<u>3985</u>	626	953	7758	2787	0.72	3.87	4.19	<u>3.69</u>	4.58
banknote	37146	552	752	<u>747</u>	714	0.05	1.40	1.21	<u>1.10</u>	1.21
landsat	1990	1669	1183	<u>1892</u>	1108	1.06	1.03	1.10	<u>1.05</u>	1.09
pendigits	7470	4037	3785	<u>4357</u>	3379	1.27	3.72	<u>2.24</u>	2.68	4.32
usps	8253	5012	395	<u>8192</u>	6374	1.05	1.93	1.96	1.43	<u>1.11</u>
letters	419	715	502	541	<u>576</u>	11.92	<u>11.01</u>	17.19	16.18	9.82
mnist	<u>52307</u>	46188	25404	81764	46269	<u>0.96</u>	1.48	1.76	1.29	0.87
skin	1279417	<u>6475</u>	3889	47859	44706	0.13	3.62	4.82	1.46	1.48
gisette_cos	21224	6171	5549	<u>19091</u>	9896	0.34	0.98	1.02	0.51	0.70
rcv1	<u>15722</u>	NA	NA	17381	13111	<u>15.73</u>	NA	NA	21.28	13.17

Dataset	Accuracy ($\uparrow$)					Normalized Mutual Information ($\uparrow$)				
	Ψ -DR	NtS	ItS	PaCMAP	LocalMAP	Ψ -DR	NtS	ItS	PaCMAP	LocalMAP
COIL20	0.997	0.878	0.868	0.875	<u>0.885</u>	0.960	0.826	0.831	<u>0.910</u>	0.9
banknote	<u>0.996</u>	<u>0.996</u>	0.993	1.000	1.000	0.936	0.603	<u>0.701</u>	0.392	0.447
landsat	<u>0.855</u>	0.848	0.853	<u>0.855</u>	0.870	<u>0.681</u>	0.633	<u>0.576</u>	0.690	0.643
pendigits	0.906	0.982	0.974	<u>0.987</u>	0.991	0.819	0.818	0.733	0.872	<u>0.832</u>
usps	0.825	0.874	0.870	<u>0.918</u>	0.952	<u>0.799</u>	0.663	0.602	0.756	0.815
letters	0.607	<u>0.797</u>	0.789	0.790	0.852	<u>0.513</u>	0.499	0.435	0.520	0.504
mnist	0.970	0.876	0.853	0.919	0.933	0.814	0.630	0.600	0.740	0.807
skin	0.996	<u>0.998</u>	0.989	<u>0.998</u>	0.999	0.828	0.086	0.047	0.022	0.035
gisette_cos	0.931	<u>0.952</u>	0.959	<u>0.975</u>	0.981	0.655	0.678	0.687	0.736	<u>0.690</u>
rcv1	0.547	NA	NA	<u>0.620</u>	0.656	<u>0.343</u>	NA	NA	0.360	0.336

Table 8. Summary of the Comparison.

	Ψ -DR	NtS	ItS	PaCMAP	LocalMAP
winner	18	2	0	9	12
runner-up	11	5	2	15	6

Table 9. Computation Time (seconds).

Datasets	Ψ -DR	NtS	ItS	PaCMAP	LocalMAP
COIL20	0.66	3.72	<u>1.35</u>	21.62	14.09
banknote	0.15	<u>1.25</u>	1.28	13.45	16.05
landsat	0.16	<u>1.16</u>	2.30	14.54	13.80
pendigits	2.19	<u>7.39</u>	7.83	20.28	22.3
usps	6.32	8.86	9.15	24.50	26.05
letters	10.47	<u>13.07</u>	14.27	32.95	33.40
mnist	28.55	<u>66.29</u>	70.52	87.28	82.90
skin	22.71	<u>174</u>	187	332	339
gisette_cos	2.92	<u>9.08</u>	9.66	26.39	25.51
rcv1	382	NA	NA	2911	<u>2809</u>

in Appendix K. The results demonstrate that only our Ψ -DR is linear with respect to both the number of points n and the original dimension d .

7. Conclusion

In this paper, we presented a novel perspective on dimensionality reduction by proposing a distributional invariant that preserves the similarity of points and distributions across the original and reduced spaces. To strictly enforce this invariant, we introduced Ψ -DR (or PSIDR), a linear-time algorithm designed for high-dimensional data. Through rigorous theoretical analysis and extensive experiments on both standard benchmarks and single-cell gene expression datasets, we demonstrated the effectiveness of our approach. Ψ -DR not only yields superior internal validation scores (CH and DB indices) but also significantly enhances performance in downstream tasks such as classification and clustering. Notably, its linear time complexity $O(nd)$ ensures scalability, making Ψ -DR a highly efficient solution for processing massive datasets with hundreds of thousands of samples and tens of thousands of dimensions.

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Impact Statements

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Notion Summary

The notion we used in the main text is shown in Table 10 and the notion we used in the proof is shown in Table 11.

Table 10. Summary of notations and their descriptions used in this paper.

Notation	Description
n	Number of data points (sample size)
d	Dimensionality of the original input space
p	Dimensionality of the target embedding space ($p \ll d$)
k	Number of underlying distributions (or clusters)
$\mathcal{X}$	Input domain, a compact subset of $\mathbb{R}^d$
x_i	The i -th data point, $x_i \in \mathcal{X}$
ν_c	The c -th underlying probability distribution
C_c	The set of indices belonging to the c -th subset
$n_{\min}$	The size of the smallest subset, $n_{\min} := \min_c C_c $
$\kappa(\cdot, \cdot)$	Continuous, positive definite kernel function
$\mathcal{H}_\kappa$	Reproducing Kernel Hilbert Space (RKHS) induced by κ
$\phi(\cdot)$	Feature map $\phi : \mathcal{X} \rightarrow \mathcal{H}_\kappa$
μ_c	True Kernel Mean Embedding (KME) of distribution ν_c
$\hat{\mu}_c$	Empirical KME estimated from subset C_c
ε	Upper bound of the kernel norm, $\sup_x \kappa(x, x) \leq \varepsilon$
z_i	Low-dimensional embedding of x_i in $\mathbb{R}^p$
$\bar{z}_c$	Low-dimensional embedding (center) of the c -th distribution
$\mathbf{Z}$	The embedding matrix of all points, $\mathbf{Z} \in \mathbb{R}^{n \times p}$
$\bar{\mathbf{Z}}$	The embedding matrix of distributional centers, $\bar{\mathbf{Z}} \in \mathbb{R}^{k \times p}$
$\mathbf{D}$	Matrix of squared MMD distances between distributions ($k \times k$)
d_{ic}	Target Euclidean distance between point i and center c derived from KME
Δ_p	Eigengap between the p -th and $(p+1)$ -th eigenvalues
λ	Regularization parameter balancing the two loss terms

B. Closed-Form Multilateration via Linearization

The localization of each point $z_i \in \mathbb{R}^p$ given the estimated distances $\{d_{ic}\}_{c=1}^k$ to the fixed anchors $\{\bar{z}_c\}_{c=1}^k$ can be formulated as finding the intersection of k hyperspheres. This requires solving the following system of quadratic equations:

$$\|z_i - \bar{z}_c\|_2^2 = d_{ic}^2, \quad c = 1, \dots, k. \quad (3)$$

Directly minimizing the non-linear least squares objective is computationally expensive and sensitive to initialization. Instead, we employ an efficient **linearized algebraic closed-form solution** (Norrdine, 2012).

Expanding the squared norm yields:

$$\|z_i\|_2^2 - 2z_i^\top \bar{z}_c + \|\bar{z}_c\|_2^2 = d_{ic}^2. \quad (4)$$

Since the term $\|z_i\|_2^2$ is unknown and non-linear, we eliminate it by selecting the first anchor $\bar{z}_1$ as a reference and subtracting its corresponding equation from the equations of the remaining anchors $c = 2, \dots, k$. The subtraction yields a set of $k-1$ linear equations:

$$2z_i^\top (\bar{z}_1 - \bar{z}_c) = d_{ic}^2 - d_{i1}^2 - \|\bar{z}_c\|_2^2 + \|\bar{z}_1\|_2^2. \quad (5)$$

This system can be written in the matrix form $\mathbf{A}z_i = \mathbf{b}_i$, where the measurement matrix $\mathbf{A} \in \mathbb{R}^{(k-1) \times p}$ and the observation vector $\mathbf{b}_i \in \mathbb{R}^{k-1}$ are defined as:

$$\mathbf{A}_{c-1} = 2(\bar{z}_1 - \bar{z}_c)^\top, \quad (6)$$

$$(\mathbf{b}_i)_{c-1} = d_{ic}^2 - d_{i1}^2 - \|\bar{z}_c\|_2^2 + \|\bar{z}_1\|_2^2, \quad (7)$$

Table 11. Additional notations and mathematical operators used in proofs.

Notation	Description
$(\cdot)^\dagger$	The Moore-Penrose pseudoinverse of a matrix
$\ \cdot\ _F$	The Frobenius norm of a matrix
$\ \cdot\ _{\text{op}}$	The spectral norm (operator norm) of a matrix
$\mathcal{O}(p)$	The set of $p \times p$ orthogonal matrices
$\mathbf{I}_k$	Identity matrix of size $k \times k$
$\mathbf{H}$	Centering matrix, $\mathbf{H} = \mathbf{I}_k - \frac{1}{k} \mathbf{1}_k \mathbf{1}_k^\top$
$\mathbf{D}^*$	The "true" squared distance matrix based on population KMEs μ_c
$\mathbf{B}, \mathbf{B}^*$	Double-centered Gram matrices derived from $\mathbf{D}$ and $\mathbf{D}^*$
$\mathbf{U}_p, \mathbf{U}_p^*$	Matrices containing the top- p eigenvectors of $\mathbf{B}$ and $\mathbf{B}^*$
$\Theta(\cdot, \cdot)$	Diagonal matrix of canonical angles between two subspaces
$\mathbf{A}$	The coefficient matrix in the linearized multilateration system
σ_0	Lower bound on the smallest singular value of $\mathbf{A}$ (conditioning parameter)
$\mathbf{b}$	The observation vector in the linearized multilateration system
z_i^*	The theoretical ideal embedding of x_i preserving true Hilbert distances
$\mathbf{Q}$	Optimal orthogonal alignment matrix used in perturbation analysis
L	Lipschitz constant for the multilateration solution
ϵ_n	Statistical error bound term, scaling with $O(1/\sqrt{n_{\min}})$

for $c = 2, \dots, k$.

The least-squares solution is then given by the normal equations:

$$z_i = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}_i. \quad (8)$$

In our implementation, we solve this utilizing singular value decomposition (SVD) for numerical stability.

Robust Fallback Strategy. In degenerate configurations where the anchors lie on a lower-dimensional manifold (e.g., collinear anchors in 2D or coplanar in 3D), the matrix $\mathbf{A}$ becomes rank-deficient or ill-conditioned. To handle such cases, we monitor the condition of the linear system. If the linear solver fails (indicating singularity), we fall back to a numerical optimization approach, minimizing the original non-linear objective function via the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm (Head & Zerner, 1985), initialized at the centroid of the anchors.

C. Proof of Lemma 5.4.

Lemma 5.4. Suppose Assumption 5.1 holds. For any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\max_{c \in [k]} \|\hat{\mu}_c - \mu_c\|_{\mathcal{H}_\kappa} \leq \sqrt{\frac{\varepsilon}{n_{\min}}} \left(1 + \sqrt{6 \log \frac{2k}{\delta}} \right).$$

Proof. Fix an arbitrary subset $c \in [k]$, and let $n_c = |C_c|$. The empirical kernel mean embedding is given by

$$\hat{\mu}_c = \frac{1}{n_c} \sum_{i \in C_c} \phi(x_i),$$

where the samples $\{x_i\}_{i \in C_c}$ are assumed to be i.i.d. draws from the distribution ν_c . By Assumption 5.1, the kernel is bounded, which implies $\|\phi(x)\|_{\mathcal{H}_\kappa}^2 = k(x, x) \leq \varepsilon$ for all $x \in \mathcal{X}$. Consequently, each summand $\phi(x_i)$ is a bounded random variable in the Hilbert space $\mathcal{H}_\kappa$ with $\|\phi(x_i)\|_{\mathcal{H}_\kappa} \leq \sqrt{\varepsilon}$ almost surely.

We apply the Upper tail of Talagrand’s inequality, Bousquet’s version (see, e.g., Theorem 3.3.9 in Giné & Nickl (2021)). For any $t > 0$,

Let

$$Z = n_c \|\hat{\mu}_c - \mu_c\|_{\mathcal{H}_\kappa} = \sup_{f \in \mathcal{H}_\kappa, \|f\| \leq 1} \sum_{i=1}^{n_c} (f(x_i) - \mathbb{E}[f(x)]),$$

for f , $\|f\|_{\mathcal{H}_\kappa} \leq 1$, we have $|f(x)| \leq \sqrt{\kappa(x, x)} \leq \sqrt{\varepsilon}$, so:

$$|f(x) - \mathbb{E}[f(x)]| \leq |f(x)| + |\mathbb{E}[f(x)]| \leq \sqrt{\varepsilon} + \sqrt{\varepsilon} = 2\sqrt{\varepsilon} = U.$$

$$\sup_{f \in \mathcal{H}_\kappa, \|f\| \leq 1} \mathbb{E}[(f(x))^2] \leq \sup_f \|f\|_\infty^2 \leq \varepsilon = \sigma^2$$

$$\mathbb{E}[Z] = n_c \mathbb{E}\|\hat{\mu}_c - \mu_c\| \leq n_c \sqrt{\mathbb{E}\|\hat{\mu}_c - \mu_c\|^2} \leq n_c \sqrt{\frac{\varepsilon}{n_c}} = \sqrt{n_c \varepsilon}$$

$$V = n_c \sigma^2 + 2U \mathbb{E}[Z] \leq n_c \varepsilon + 2(2\sqrt{\varepsilon})(\sqrt{n_c \varepsilon}) = n_c \varepsilon + 4\varepsilon \sqrt{n_c} \leq 2n_c \varepsilon. (n_c \geq 16)$$

this yields

$$\mathbb{P}\left(\|\hat{\mu}_c - \mu_c\|_{\mathcal{H}_\kappa} \geq \sqrt{\frac{\varepsilon}{n_c}} + t\right) \leq \exp\left(-\frac{n_c t^2}{4\varepsilon + \frac{4}{3}\sqrt{\varepsilon}t}\right).$$

For $t \leq \sqrt{\varepsilon}$, the denominator is at most 6ε , so the bound simplifies to

$$\mathbb{P}\left(\|\hat{\mu}_c - \mu_c\|_{\mathcal{H}_\kappa} \geq \sqrt{\frac{\varepsilon}{n_c}} + t\right) \leq \exp\left(-\frac{n_c t^2}{6\varepsilon}\right).$$

To obtain a high-probability bound, we set the right-hand side equal to $\delta/(2k)$. Solving for t gives

$$t = \sqrt{\frac{6\varepsilon}{n_c} \log \frac{2k}{\delta}}.$$

Thus, with probability at least $1 - \delta/(2k)$,

$$\|\hat{\mu}_c - \mu_c\|_{\mathcal{H}_\kappa} \leq \sqrt{\frac{\varepsilon}{n_c}} + \sqrt{\frac{6\varepsilon}{n_c} \log \frac{2k}{\delta}} = \sqrt{\frac{\varepsilon}{n_c}} \left(1 + \sqrt{6 \log \frac{2k}{\delta}}\right).$$

Since $n_c \geq n_{\min}$ by definition of $n_{\min}$, and the function $x \mapsto 1/\sqrt{x}$ is decreasing, we have $\sqrt{1/n_c} \leq \sqrt{1/n_{\min}}$. Therefore, the bound holds with $n_{\min}$ in place of n_c :

$$\|\hat{\mu}_c - \mu_c\|_{\mathcal{H}_\kappa} \leq \sqrt{\frac{\varepsilon}{n_{\min}}} \left(1 + \sqrt{6 \log \frac{2k}{\delta}}\right).$$

This bound holds for a fixed subset c . To obtain a uniform bound over all k subsets, we apply the union bound. Specifically, let E_c denote the event that the above inequality fails for subset c . Then $\mathbb{P}(E_c) \leq \delta/(2k)$, and

$$\mathbb{P}\left(\bigcup_{c=1}^k E_c\right) \leq \sum_{c=1}^k \mathbb{P}(E_c) \leq k \cdot \frac{\delta}{2k} = \frac{\delta}{2} < \delta.$$

Thus, with probability at least $1 - \delta$, the inequality holds simultaneously for all $c \in [k]$, which implies

$$\max_{c \in [k]} \|\hat{\mu}_c - \mu_c\|_{\mathcal{H}_\kappa} \leq \sqrt{\frac{\varepsilon}{n_{\min}}} \left(1 + \sqrt{6 \log \frac{2k}{\delta}}\right).$$

□

D. Proof of Lemma 5.5.

Lemma 5.5. Let $\mathbf{D}, \mathbf{D}^* \in \mathbb{R}^{k \times k}$ be Euclidean distance matrices, and let $\bar{\mathbf{Z}}, \bar{\mathbf{Z}}^* \in \mathbb{R}^{k \times p}$ be their classical MDS embeddings (i.e., $\bar{\mathbf{Z}}\bar{\mathbf{Z}}^\top = \mathbf{B}$, where $\mathbf{B} = -\frac{1}{2}\mathbf{H}\mathbf{D}\mathbf{H}$, and similarly for $\bar{\mathbf{Z}}^*$). Let $\mathcal{O}(p)$ denote the set of $p \times p$ orthogonal matrices. Under Assumption 3,

$$\min_{\mathbf{Q} \in \mathcal{O}(p)} \|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}^* \mathbf{Q}\|_F \leq \frac{\sqrt{2p}}{2\Delta_p} \|\mathbf{D} - \mathbf{D}^*\|_F.$$

Proof. Let $\mathbf{H} = \mathbf{I}_k - \frac{1}{k}\mathbf{1}_k\mathbf{1}_k^\top$ be the centering matrix. The Gram matrices corresponding to $\mathbf{D}$ and $\mathbf{D}^*$ are

$$\mathbf{B} = -\frac{1}{2}\mathbf{H}\mathbf{D}\mathbf{H}, \quad \mathbf{B}^* = -\frac{1}{2}\mathbf{H}\mathbf{D}^*\mathbf{H}.$$

Their difference satisfies

$$\mathbf{B} - \mathbf{B}^* = -\frac{1}{2}\mathbf{H}(\mathbf{D} - \mathbf{D}^*)\mathbf{H}.$$

Since $\mathbf{H}$ is symmetric and idempotent ($\mathbf{H}^\top = \mathbf{H}$, $\mathbf{H}^2 = \mathbf{H}$), its spectral norm is $\|\mathbf{H}\|_2 = 1$. Using the submultiplicativity of the Frobenius norm with respect to the spectral norm, we obtain

$$\|\mathbf{B} - \mathbf{B}^*\|_F = \frac{1}{2}\|\mathbf{H}(\mathbf{D} - \mathbf{D}^*)\mathbf{H}\|_F \leq \frac{1}{2}\|\mathbf{H}\|_2\|\mathbf{D} - \mathbf{D}^*\|_F\|\mathbf{H}\|_2 = \frac{1}{2}\|\mathbf{D} - \mathbf{D}^*\|_F.$$

Let $\mathbf{U}_p, \mathbf{U}_p^* \in \mathbb{R}^{k \times p}$ be the matrices whose columns are the top- p eigenvectors of $\mathbf{B}$ and $\mathbf{B}^*$, respectively. Under Assumption 3, the eigengap $\Delta_p = \lambda_p^* - \lambda_{p+1}^*$ is positive. By the Davis–Kahan sin Θ theorem (Bhatia (1997) Theorem VII.3.1), the Frobenius norm of the difference between the orthogonal projectors onto the top- p eigenspaces satisfies

$$\|\mathbf{U}_p\mathbf{U}_p^\top - \mathbf{U}_p^*\mathbf{U}_p^{*\top}\|_F = \|\sin \Theta(\mathbf{U}_p, \mathbf{U}_p^*)\|_F \leq \sqrt{p} \|\sin \Theta(\mathbf{U}_p, \mathbf{U}_p^*)\|_{\text{op}} \leq \sqrt{p} \frac{\|\mathbf{B} - \mathbf{B}^*\|_{\text{op}}}{\Delta_p}.$$

Finally, using the general norm inequality $\|\mathbf{B} - \mathbf{B}^*\|_{\text{op}} \leq \|\mathbf{B} - \mathbf{B}^*\|_F$, we obtain the stated result:

$$\|\mathbf{U}_p\mathbf{U}_p^\top - \mathbf{U}_p^*\mathbf{U}_p^{*\top}\|_F \leq \frac{\sqrt{p}\|\mathbf{B} - \mathbf{B}^*\|_F}{\Delta_p}.$$

The classical MDS embeddings are defined as $\bar{\mathbf{Z}} = \mathbf{U}_p\mathbf{\Lambda}_p^{1/2}$ and $\bar{\mathbf{Z}}^* = \mathbf{U}_p^*\mathbf{\Lambda}_p^{*1/2}$, where $\mathbf{\Lambda}_p, \mathbf{\Lambda}_p^*$ are the diagonal matrices of the top- p eigenvalues. By Theorem 2 of Yu et al. (2015), which bounds the perturbation of low-rank factorizations under eigengap conditions, we have

$$\min_{\mathbf{Q} \in \mathcal{O}(p)} \|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}^* \mathbf{Q}\|_F \leq \sqrt{2} \|\mathbf{U}_p\mathbf{U}_p^\top - \mathbf{U}_p^*\mathbf{U}_p^{*\top}\|_F \leq \frac{\sqrt{2p}}{\Delta_p} \|\mathbf{B} - \mathbf{B}^*\|_F.$$

Substituting the bound $\|\mathbf{B} - \mathbf{B}^*\|_F \leq \frac{1}{2}\|\mathbf{D} - \mathbf{D}^*\|_F$ into the above inequality yields

$$\min_{\mathbf{Q} \in \mathcal{O}(p)} \|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}^* \mathbf{Q}\|_F \leq \frac{\sqrt{2p}}{2\Delta_p} \|\mathbf{D} - \mathbf{D}^*\|_F.$$

□

E. Proof of Lemma 5.6.

Lemma 5.6. Let $\bar{\mathbf{Z}} = [\bar{z}_1, \dots, \bar{z}_k]^\top \in \mathbb{R}^{k \times p}$ have full column rank, and define $\mathbf{A} = -2(\bar{\mathbf{Z}}_{2:k} - \mathbf{1}_{k-1}\bar{z}_1^\top) \in \mathbb{R}^{(k-1) \times p}$. Assume $\sigma_{\min}(\mathbf{A}) \geq \sigma_0 > 0$. Let $\mathbf{d} \in \mathbb{R}^k$ be distances, and define $\mathbf{b} \in \mathbb{R}^{k-1}$ by $b_c = d_{c+1}^2 - d_1^2 - (\|\bar{z}_{c+1}\|_2^2 - \|\bar{z}_1\|_2^2)$. Let $z^\dagger = \mathbf{A}^\dagger \mathbf{b}$. Then for any two inputs $(\mathbf{d}, \bar{\mathbf{Z}})$ and $(\mathbf{d}', \bar{\mathbf{Z}}')$,

$$\|z^\dagger - z'^\dagger\|_2 \leq L \left(\|\mathbf{d}^2 - \mathbf{d}'^2\|_2 + \|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}'\|_F \right),$$

where the Lipschitz constant satisfies $L = O\left(\frac{1}{\sigma_0^2} \left(1 + \frac{\|\mathbf{b}\|_2}{\sigma_0}\right)\right)$.

Proof. The multilateration solution $z^\dagger$ satisfies the normal equations $\mathbf{A}^\top \mathbf{A} z^\dagger = \mathbf{A}^\top \mathbf{b}$. Consider perturbed inputs $(\mathbf{d}', \bar{\mathbf{Z}}')$ yielding $\mathbf{A}'$ and $\mathbf{b}'$, with solution $z'^\dagger$ satisfying $\mathbf{A}'^\top \mathbf{A}' z'^\dagger = \mathbf{A}'^\top \mathbf{b}'$.

Subtracting the two equations gives

$$\mathbf{A}^\top \mathbf{A} (z^\dagger - z'^\dagger) = \mathbf{A}'^\top \mathbf{b}' - \mathbf{A}^\top \mathbf{b} - (\mathbf{A}'^\top \mathbf{A}' - \mathbf{A}^\top \mathbf{A}) z'^\dagger.$$

Let $\Delta \mathbf{A} = \mathbf{A}' - \mathbf{A}$ and $\Delta \mathbf{b} = \mathbf{b}' - \mathbf{b}$. Then

$$\mathbf{A}'^\top \mathbf{b}' - \mathbf{A}^\top \mathbf{b} = \mathbf{A}^\top \Delta \mathbf{b} + \Delta \mathbf{A}^\top \mathbf{b} + \Delta \mathbf{A}^\top \Delta \mathbf{b},$$

$$\mathbf{A}'^\top \mathbf{A}' - \mathbf{A}^\top \mathbf{A} = \mathbf{A}^\top \Delta \mathbf{A} + \Delta \mathbf{A}^\top \mathbf{A} + \Delta \mathbf{A}^\top \Delta \mathbf{A}.$$

For sufficiently small perturbations, higher-order terms $\Delta \mathbf{A}^\top \Delta \mathbf{b}$ and $\Delta \mathbf{A}^\top \Delta \mathbf{A}$ are negligible. Keeping only first-order terms,

$$\mathbf{A}^\top \mathbf{A} (z^\dagger - z'^\dagger) \approx \mathbf{A}^\top \Delta \mathbf{b} + \Delta \mathbf{A}^\top \mathbf{b} - (\mathbf{A}^\top \Delta \mathbf{A} + \Delta \mathbf{A}^\top \mathbf{A}) z'^\dagger.$$

Taking norms and using $\|\mathbf{A}^\top \mathbf{A}\|_2 \geq \sigma_0^2$,

$$\sigma_0^2 \|z^\dagger - z'^\dagger\|_2 \leq \|\mathbf{A}\|_2 \|\Delta \mathbf{b}\|_2 + \|\Delta \mathbf{A}\|_2 \|\mathbf{b}\|_2 + 2\|\mathbf{A}\|_2 \|\Delta \mathbf{A}\|_2 \|z'^\dagger\|_2. \quad (9)$$

We now derive explicit bounds for each term on the right-hand side.

Recall that $\mathbf{A} = -2\mathbf{L}\bar{\mathbf{Z}}$, where $\mathbf{L}$ is a differencing matrix with spectral norm $\|\mathbf{L}\|_2 = \sqrt{k}$. Thus,

$$\|\mathbf{A}\|_2 \leq 2\|\mathbf{L}\|_2 \|\bar{\mathbf{Z}}\|_2 = 2\sqrt{k} \|\bar{\mathbf{Z}}\|_2.$$

By linearity, $\Delta \mathbf{A} = -2\mathbf{L}(\bar{\mathbf{Z}}' - \bar{\mathbf{Z}})$. Using the fact that the spectral norm is bounded by the Frobenius norm, we have

$$\|\Delta \mathbf{A}\|_2 \leq 2\sqrt{k} \|\bar{\mathbf{Z}}' - \bar{\mathbf{Z}}\|_2 \leq 2\sqrt{k} \|\bar{\mathbf{Z}}' - \bar{\mathbf{Z}}\|_F.$$

The vector $\mathbf{b}$ is defined component-wise as $b_c = (d_{c+1}^2 - d_1^2) - (\|\bar{z}_{c+1}\|_2^2 - \|\bar{z}_1\|_2^2)$. The perturbation $\Delta \mathbf{b}$ splits into a distance part and an anchor coordinate part:

$$\|\Delta \mathbf{b}\|_2 \leq \|\Delta \mathbf{b}_{\text{dist}}\|_2 + \|\Delta \mathbf{b}_{\text{coord}}\|_2.$$

For the distance part, each component is a difference of two squared distances. The norm is bounded by:

$$\|\Delta \mathbf{b}_{\text{dist}}\|_2 \leq 2\|\mathbf{d}^2 - \mathbf{d}'^2\|_2.$$

For the coordinate part, the function $f(z) = \|z\|^2$ is locally Lipschitz with constant $2\|z\|$. The term $\|\bar{z}_{c+1}\|^2 - \|\bar{z}_1\|^2$ involves two such terms, leading to a Lipschitz constant bounded by $4 \max_c \|\bar{z}_c\|_2$. Thus:

$$\|\Delta \mathbf{b}_{\text{coord}}\|_2 \leq 4 \max_c \|\bar{z}_c\|_2 \|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}'\|_F.$$

Combining these, we obtain:

$$\|\Delta \mathbf{b}\|_2 \leq 2\|\mathbf{d}^2 - \mathbf{d}'^2\|_2 + 4 \max_c \|\bar{z}_c\|_2 \|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}'\|_F.$$

Substituting the bounds from Steps 1 and 2, and using $\|z^{\dagger'}\|_2 \lesssim \|\mathbf{b}\|_2/\sigma_0$ ($\|z^{\dagger'}\|_2 \leq \|\mathbf{A}^\dagger\|_2\|\mathbf{b}'\|_2$ neglecting higher-order terms), into (9):

$$\begin{aligned} \sigma_0^2 \|z^\dagger - z^{\dagger'}\|_2 &\leq \underbrace{\left(2\sqrt{k}\|\bar{\mathbf{Z}}\|_2\right)}_{\|\mathbf{A}\|_2} \left(2\|\mathbf{d}^2 - \mathbf{d}'^2\|_2 + 4\max_c \|\bar{z}_c\|_2 \|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}'\|_F\right) \\ &\quad + \underbrace{\left(2\sqrt{k}\|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}'\|_F\right)}_{\|\Delta\mathbf{A}\|_2} \|\mathbf{b}\|_2 \\ &\quad + 2 \underbrace{\left(2\sqrt{k}\|\bar{\mathbf{Z}}\|_2\right)}_{\|\mathbf{A}\|_2} \underbrace{\left(2\sqrt{k}\|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}'\|_F\right)}_{\|\Delta\mathbf{A}\|_2} \frac{\|\mathbf{b}\|_2}{\sigma_0}. \end{aligned}$$

Now, we strictly group the terms by the perturbation sources $\|\mathbf{d}^2 - \mathbf{d}'^2\|_2$ and $\|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}'\|_F$.

The coefficient for $\|\mathbf{d}^2 - \mathbf{d}'^2\|_2$ is:

$$C_1 = 4\sqrt{k}\|\bar{\mathbf{Z}}\|_2.$$

The coefficient for $\|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}'\|_F$ is sum of three terms:

$$C_2 = \left(8\sqrt{k}\|\bar{\mathbf{Z}}\|_2 \max_c \|\bar{z}_c\|_2\right) + \left(2\sqrt{k}\|\mathbf{b}\|_2\right) + \left(\frac{8k\|\bar{\mathbf{Z}}\|_2\|\mathbf{b}\|_2}{\sigma_0}\right).$$

Dividing the entire inequality by σ_0^2 , we get:

$$\|z^\dagger - z^{\dagger'}\|_2 \leq \frac{C_1}{\sigma_0^2} \|\mathbf{d}^2 - \mathbf{d}'^2\|_2 + \frac{C_2}{\sigma_0^2} \|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}'\|_F.$$

Finally, define the global Lipschitz constant L as the maximum of the two coefficients:

$$L = \max\left(\frac{C_1}{\sigma_0^2}, \frac{C_2}{\sigma_0^2}\right) = O\left(\frac{1}{\sigma_0^2} \left(1 + \frac{\|\mathbf{b}\|_2}{\sigma_0}\right)\right).$$

This yields the desired form:

$$\|z^\dagger - z^{\dagger'}\|_2 \leq L \left(\|\mathbf{d}^2 - \mathbf{d}'^2\|_2 + \|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}'\|_F\right).$$

□

F. Proof of Theorem 5.7.

Theorem 5.7. Let $z_i^* \in \mathbb{R}^p$ satisfy $\|z_i^* - \bar{z}_c^*\|_2^2 = 2(1 - \langle \phi(x_i), \mu_c \rangle_{\mathcal{H}_\kappa})$ for all c . Let z_i be the embedding from Algorithm 1. Then, as $n \rightarrow \infty$,

$$\frac{1}{n} \sum_{i=1}^n \|z_i - z_i^*\|_2^2 \xrightarrow{\mathbb{P}} 0.$$

Proof. By Lemma 5.4, with probability $\geq 1 - \delta$, $\max_c \|\hat{\mu}_c - \mu_c\|_{\mathcal{H}_\kappa} \leq \epsilon_n := \sqrt{\frac{\epsilon}{n_{\min}}} (1 + \sqrt{6 \log \frac{2k}{\delta}})$.

Let $d_{ic}^2 = \|\phi(x_i) - \hat{\mu}_c\|_{\mathcal{H}_\kappa}^2$ and $(d_{ic}^*)^2 = \|\phi(x_i) - \mu_c\|_{\mathcal{H}_\kappa}^2$.

Using the algebraic identity $|a^2 - b^2| = |a - b| \cdot |a + b|$:

$$|d_{ic}^2 - (d_{ic}^*)^2| \leq \underbrace{\|(\phi(x_i) - \hat{\mu}_c) - (\phi(x_i) - \mu_c)\|_{\mathcal{H}_\kappa}}_{\text{Error term}} \cdot \underbrace{(\|\phi(x_i) - \hat{\mu}_c\|_{\mathcal{H}_\kappa} + \|\phi(x_i) - \mu_c\|_{\mathcal{H}_\kappa})}_{\text{Scale term}}.$$

The error term simplifies to $\|\hat{\mu}_c - \mu_c\|_{\mathcal{H}_\kappa} \leq \epsilon_n$. The scale term is bounded by $2\sqrt{\epsilon} + 2\sqrt{\epsilon} = 4\sqrt{\epsilon}$. Thus, $|d_{ic}^2 - (d_{ic}^*)^2| \leq 4\sqrt{\epsilon}\epsilon_n$.

$$|\mathbf{d}_i^2 - (\mathbf{d}_i^*)^2|_2 = \sqrt{\sum c = 1^k |d_{ic}^2 - (d_{ic}^*)^2|^2} \leq \sqrt{k \cdot (4\sqrt{\varepsilon}\epsilon_n)^2} = 4\epsilon_n \sqrt{k\varepsilon}. \quad (10)$$

Similarly, for the MMD matrix, let $D_{cc'} = \|\hat{\mu}_c - \hat{\mu}_{c'}\|^2$ and $D_{cc'}^* = \|\mu_c - \mu_{c'}\|^2$. The error term is bounded by $\|\hat{\mu}_c - \mu_c\| + \|\hat{\mu}_{c'} - \mu_{c'}\| \leq 2\epsilon_n$. The scale term is bounded by $4\sqrt{\varepsilon}$. Thus, $|D_{cc'} - D_{cc'}^*| \leq (2\epsilon_n)(4\sqrt{\varepsilon}) = 8\sqrt{\varepsilon}\epsilon_n$.

Converting to the Frobenius norm for the $k \times k$ matrix:

$$|\mathbf{D} - \mathbf{D}^*|_F = \sqrt{\sum c, c' |D_{cc'} - D_{cc'}^*|^2} \leq \sqrt{k^2 \cdot (8\sqrt{\varepsilon}\epsilon_n)^2} = k \cdot 8\sqrt{\varepsilon}\epsilon_n. \quad (11)$$

By Lemma 5.5, $\min_{\mathbf{Q}} \|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}^* \mathbf{Q}\|_F \leq C_1 k \epsilon_n$ for $C_1 = 4\sqrt{2p\varepsilon}/\Delta_p$.

Lemma 5.6 then gives

$$\|z_i - z_i^* \mathbf{Q}\|_2 \leq L (\|\mathbf{d}_i^2 - (\mathbf{d}_i^*)^2\|_2 + \|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}^* \mathbf{Q}\|_F).$$

So:

$$\|z_i - z_i^* \mathbf{Q}\|_2 \leq L \left(\underbrace{\sqrt{k} C_2 \epsilon_n}_{\text{Distance term}} + \underbrace{C_1 k \epsilon_n}_{\text{Anchor term}} \right).$$

where $C_2 = 4\sqrt{\varepsilon}$

Let $C_{\text{total}} = L(C_2 + C_1)$. Thus: $\|z_i - z_i^* \mathbf{Q}\|_2 \leq C_{\text{total}} \cdot k \epsilon_n$.

Finally, we consider the average squared error. Squaring the inequality:

$$\|z_i - z_i^* \mathbf{Q}\|_2^2 \leq C_{\text{total}}^2 k^2 \epsilon_n^2.$$

Averaging over all n points:

$$\frac{1}{n} \sum_{i=1}^n \|z_i - z_i^* \mathbf{Q}\|_2^2 \leq C_{\text{total}}^2 k^2 \epsilon_n^2.$$

Recall that $\epsilon_n = O(1/\sqrt{n_{\min}})$. Under Assumption 5.2, $n_{\min} \rightarrow \infty$ as $n \rightarrow \infty$. Therefore, $\epsilon_n \rightarrow 0$, which implies that the mean squared error converges to 0 in probability. $\square$

G. Proof of Theorem 5.8.

Theorem 5.8. Under Assumptions 5.1, 5.2, and 5.3, there exists a constant $C > 0$, such that for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\frac{1}{n} \sum_{i=1}^n \|z_i - z_i^* \mathbf{Q}\|_2^2 \leq C \cdot \frac{k^2}{n_{\min}} \left(1 + \log \frac{2k}{\delta} \right).$$

Proof. From the proof of Theorem 5.7, there exists a constant $C_{\text{total}} > 0$ such that for all i , with probability $\geq 1 - \delta$,

$$\|z_i - z_i^* \mathbf{Q}\|_2 \leq C_{\text{total}} k \epsilon_n, \quad \text{where} \quad \epsilon_n = \sqrt{\frac{\varepsilon}{n_{\min}}} \left(1 + \sqrt{6 \log \frac{2k}{\delta}} \right).$$

Squaring both sides and averaging over i gives

$$\frac{1}{n} \sum_{i=1}^n \|z_i - z_i^* \mathbf{Q}\|_2^2 = \frac{1}{n} \sum_{i=1}^n \|z_i - z_i^* \mathbf{Q}\|_2^2 \leq C_{\text{total}}^2 k^2 \epsilon_n^2.$$

Using the inequality $(1 + \sqrt{a})^2 \leq 2(1 + a)$ for $a \geq 0$, we obtain

$$\epsilon_n^2 \leq \frac{2\varepsilon}{n_{\min}} \left(1 + 6 \log \frac{2k}{\delta} \right) \leq \frac{12}{n_{\min}} \left(1 + \log \frac{2k}{\delta} \right).$$

Setting $C = 12\varepsilon C_{\text{total}}^2$ completes the proof. $\square$

H. Proof of Theorem 5.9.

Theorem 5.9. Assume the kernel is normalized ($k(x, x) = 1$ for all x). Let $d_{ic}^{\mathcal{H}} = \|\phi(x_i) - \mu_c\|_{\mathcal{H}_\kappa}$ and $d_{ic}^{\mathbb{R}^p} = \|z_i - \bar{z}_c\|_2$. Under Assumptions 5.1, 5.2, and 5.3, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\max_{i \in [n], c \in [k]} \left| d_{ic}^{\mathbb{R}^p} - d_{ic}^{\mathcal{H}} \right| \leq C' \cdot \frac{k}{\sqrt{n_{\min}}} \left(1 + \sqrt{\log \frac{2k}{\delta}} \right),$$

for some constant $C' > 0$.

Proof. We aim to bound the difference between the recovered Euclidean distance and the true Hilbert space distance:

$$\Delta_{ic} = \left| d_{ic}^{\mathbb{R}^p} - d_{ic}^{\mathcal{H}} \right| = \left| \|z_i - \bar{z}_c\|_2 - \|\phi(x_i) - \mu_c\|_{\mathcal{H}_\kappa} \right|.$$

Recall from Theorem 5.7 that the "true" embeddings z_i^* and centers $\bar{z}_c^*$ are defined to perfectly preserve the Hilbert space distances. That is, for any rotation matrix $\mathbf{Q} \in \mathcal{O}(p)$,

$$\|\phi(x_i) - \mu_c\|_{\mathcal{H}_\kappa} = \|z_i^* - \bar{z}_c^*\|_2 = \|z_i^* \mathbf{Q} - \bar{z}_c^* \mathbf{Q}\|_2.$$

Substituting this into the expression for Δ_{ic} :

$$\Delta_{ic} = \left| \|z_i - \bar{z}_c\|_2 - \|z_i^* \mathbf{Q} - \bar{z}_c^* \mathbf{Q}\|_2 \right|.$$

Using the reverse triangle inequality $|||u|| - ||v||| \leq ||u - v||$, we have:

$$\Delta_{ic} \leq \|(z_i - \bar{z}_c) - (z_i^* \mathbf{Q} - \bar{z}_c^* \mathbf{Q})\|_2.$$

By the standard triangle inequality, we can separate the error contributions from the point z_i and the center $\bar{z}_c$:

$$\Delta_{ic} \leq \|z_i - z_i^* \mathbf{Q}\|_2 + \|\bar{z}_c - \bar{z}_c^* \mathbf{Q}\|_2.$$

We now bound each term using our previous results. Let $\mathbf{Q}$ be the optimal alignment matrix from Lemma 5.5.

From Lemma 5.5 and the derivation in Theorem 5.7, the MDS centers satisfy:

$$\|\bar{z}_c - \bar{z}_c^* \mathbf{Q}\|_2 \leq \|\bar{\mathbf{Z}} - \bar{\mathbf{Z}}^* \mathbf{Q}\|_F \leq C_1 k \epsilon_n.$$

From the proof of Theorem 5.7, the multilateration solution satisfies:

$$\|z_i - z_i^* \mathbf{Q}\|_2 \leq C_{\text{total}} k \epsilon_n.$$

Summing these bounds:

$$\Delta_{ic} \leq (C_{\text{total}} + C_1) k \epsilon_n.$$

Recall that $\epsilon_n = \sqrt{\frac{\varepsilon}{n_{\min}}} \left(1 + \sqrt{6 \log \frac{2k}{\delta}} \right)$. Defining a new constant $C' = C_{\text{total}} + C_1$, and noting that $1 + \sqrt{6X} \leq \sqrt{6}(1 + \sqrt{X})$ allows us to absorb constants for a cleaner form, we obtain:

$$\max_{i,c} \left| d_{ic}^{\mathbb{R}^p} - d_{ic}^{\mathcal{H}} \right| \leq C' \cdot \frac{k}{\sqrt{n_{\min}}} \left(1 + \sqrt{\log \frac{2k}{\delta}} \right).$$

□

I. Limitation

The performance of Ψ -DR is determined by the ‘weakest link’, the least represented distribution. To achieve a low embedding error, the dataset should ideally be balanced; for highly imbalanced data, n_{min} becomes the bottleneck.

J. Experiments Setting

The Characteristics of the datasets used are shown in Table 12. Rcv1 is a multi-label dataset, and we selected its first label.

Table 12. Characteristics of the datasets used.

Datasets	#points	#dimensions	k
COIL20	1440	1024	20
Banknote	1372	4	2
Landsat	2000	36	6
Pendigits	10992	16	10
USPS	11000	256	10
Letters	20000	16	26
MNIST	70000	784	10
Skin	245057	3	2
Gisette	7000	5000	2
Rcv1	804414	47236	37

The Characteristics of the single-cell expression datasets used are shown in Table 13. We use the highly variable gene (HVG) selection (Zhao et al., 2025) select 2000 genes before dimensionality reduction. And we set k be the number of types. We employed the isolation kernel (Ting et al., 2018) that was used in KBC in our experiments.

Table 13. Characteristics of the single-cell expression datasets used.

Datasets	#Cells	#Genes	#Types	Accession	Description
airway	7193	27716	7	GSE103354	mouse tracheal epithelium cells
crohn	39563	33660	27	Broad Institute Single Cell Portal	human uninflamed ileum
tonsil	5778	36601	13	Broad Institute Single Cell Portal	human tonsil
jurkat	3234	32738	2	10X genomics	artificially planted cells

K. Scale-up test

The results of the runtime are shown in Table 14. Our method is linear with respect to both number of points n and dimensions d , while Neg-t-SNE and InfoNC-t-SNE are linear with respect to dimensions d but have greater than linear time complexity with respect to the number of points n . PaCMAP and LocalMAP also have greater than linear time complexity with respect to the number of points n , but they are less affected by the number of dimensions d .

L. Ablation Study

Our contribution is proposing a point-distribution similarity invariant, so we conducted the following ablation experiments. We compared the results of maintaining this invariant (Ψ -DR) with maintaining the point-point similarity invariant (PP-DR). The point-point similarity invariant reduces dimensionality by preserving the distance between each point in C_c (the second line of the algorithm) and the center of C_c . The results are shown in Table 15. The results demonstrate that our proposed point-distribution similarity invariant is effective.

Table 14. Runtime and Scale-up test.

		Time					Ratio					
		Ψ -DR	NtS	ItS	PaCMAP	LocalMAP	n	Ψ -DR	NtS	ItS	PaCMAP	LocalMAP
n	10000	5.84	51.23	36.32	11.32	11.38	1	1.00	1.00	1.00	1.00	1.00
	20000	8.94	106	108	25.10	25.48	2	1.53	2.07	2.99	2.22	2.24
	50000	23.39	644	648	67.49	68.07	5	4.01	12.57	17.85	5.96	5.98
	100000	40.92	2049	2043	161	159	10	7.01	40.01	56.26	14.19	13.97
	200000	89.31			387	399	20	15.29	0.00	0.00	34.21	35.04
	500000	225			1519	1492	50	38.61	0.00	0.00	134.22	131.09
d	1000	16.13	3080	3131	1415	1510	1	1.00	1.00	1.00	1.00	1.00
	2000	23.22	5610	5665	1439	1537	2	1.44	1.82	1.81	1.02	1.02
	5000	45.99	13186	13240	1506	1678	5	2.85	4.28	4.23	1.06	1.11
	10000	84.25			1585	1698	10	5.22	0.00	0.00	1.12	1.12
	20000	169			1675	1714	20	10.53	0.00	0.00	1.18	1.14
	40000	320			2548	2043	40	19.84	0.00	0.00	1.80	1.35

Table 15. Ablation study.

Datasets	CH		DB		NMI		ACC	
	Ψ -DR	PP-DR	Ψ -DR	PP-DR	Ψ -DR	PP-DR	Ψ -DR	PP-DR
COIL20	3985	3178	0.72	1.20	0.960	0.927	0.997	0.979
banknote	37146	108	0.05	2.18	0.936	0.120	0.996	0.585
landsat	1990	1617	1.06	1.06	0.681	0.675	0.855	0.85
pendigits	7470	4651	1.27	2.00	0.819	0.791	0.906	0.905
usps	8253	6421	1.05	8.59	0.799	0.802	0.825	0.81
letters	418.739	509	11.917	11.45	0.513	0.484	0.607	0.576
mnist	52307	48181	0.96	1.64	0.814	0.815	0.97	0.841
skin	1279417	10467	0.13	3.47	0.828	0.011	0.996	0.984
gisette_cos	21224	655	0.34	3.06	0.655	0.063	0.931	0.596
rcv1	15722	4210	15.73	21.23	0.343	0.348	0.547	0.499